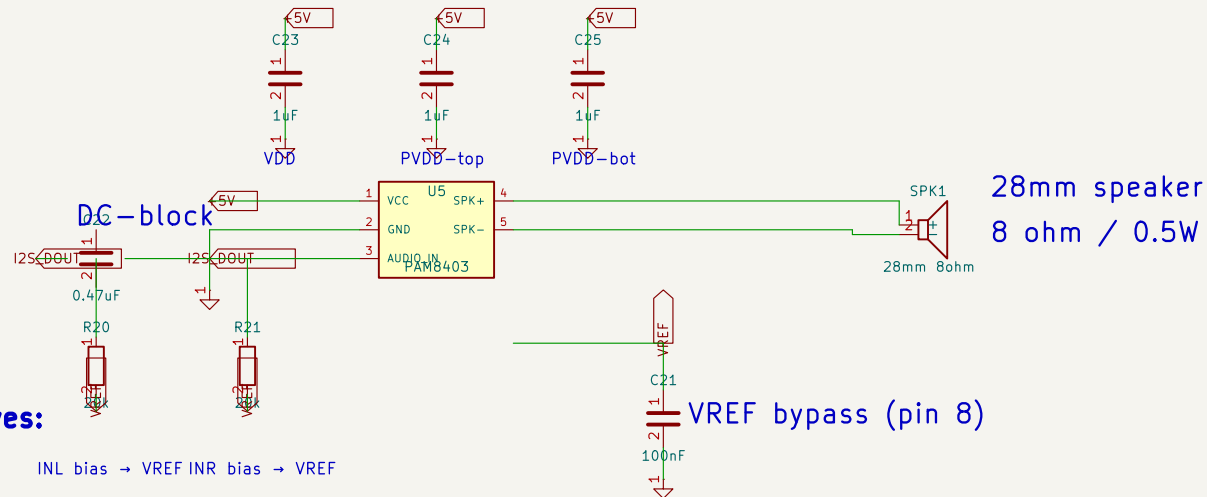


AUDIO OUTPUT

I2S DAC → PAM8403 Class-D Amplifier → 28mm 8ohm Speaker

I2S Bus (PDM TX mode – DOUT only):

I2S_BCLK	GPIO15 – Bit Clock (UNUSED – PDM mode, v2 free)
I2S_LRCK	GPIO16 – L/R Word Select (UNUSED – PDM mode, v2 free)
I2S_DOUT	GPIO17 – PDM sigma-delta out @ 32kHz → PAM8403



Design Notes:

- PAM8403: filterless Class-D stereo amplifier (3W per channel max)
- Only one channel used for mono audio
- No external DAC needed: ESP32-S3 has built-in I2S with DMA
- Powered from +5V rail for max headroom (3.3V limits volume)
- DMA-driven audio streaming for low CPU overhead